Antonie van Leeuwenhoek (2021) 114:2205–2217
https://doi.org/10.1007/s10482-021-01673-w

ORIGINAL PAPER

Genome analysis of alginate synthesizing *Pseudomonas aeruginosa* strain SW1 isolated from degraded seaweeds

Rajathirajan Siva Dharshini · Ranjani Manickam · Wayne R. Curtis · Pasupathi Rathinasabapathi · Mohandass Ramya

Received: 11 August 2021 / Accepted: 7 October 2021 / Published online: 18 October 2021

Abstract *Pseudomonas aeruginosa* strain SW1 is an aerobic, motile, Gram-negative, and rod-shaped bacterium isolated from degraded seaweeds. Based on the 16S rRNA gene sequence and MALDI TOF analysis, strain SW1 exhibits 100% similarity to *P. aeruginosa* DSM 50,071, its closest phylogenetic neighbor. The complete genome of strain SW1 consists of a single circular chromosome with 23,258,857 bp (G + C content of 66%), including 6734 protein-coding sequences, 8 rRNA, and 63 tRNA sequences. The genome of the *P. aeruginosa* SW1 contains at least 27 genes for the biosynthesis of alginate and other exopolysaccharide involved in biofilm formation. KAAS and GO analysis and functional annotation by COG and CAZymes are consistent with the biosynthesis of alginate. In addition, the presence of antimicrobial resistance, multi-efflux operon, and antibiotic inactivation genes indicate a pathogenic potential similar to strain DSM50071. The high-quality genome and associated annotation provide a starting point to exploit the potential for *P. aeruginosa* to produce alginate.

Keywords *Pseudomonas aeruginosa* · Alginate biosynthesis · Antibiotic microbial resistance · Seaweeds · Biofilm formation

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10482-021-01673-w.

R. S. Dharshini · P. Rathinasabapathi · M. Ramya (✉)
Faculty of Engineering and Technology, Molecular Genetics Laboratory, Department of Genetic Engineering, School of Bioengineering, SRM Institute of Science and Technology, Kattankulathur, Chengalpattu, Tamil Nadu 603203, India
e-mail: ramya.mohandass@gmail.com

R. Manickam
SRM-DBT Platform for Advanced Life Science Technologies, SRMIST, Chengalpattu, Chennai, Tamil Nadu 603203, India

W. R. Curtis
Department of Chemical Engineering, The Pennsylvania State University, University Park, PA, USA

Introduction

Alginate is the major structural component of seaweeds that provides mechanical strength and flexibility to withstand the harsh marine environment. Many bacteria can produce alginate, a linear copolymer composed of uronic acids such as alpha L–guluronic acid and beta D- mannuronic acid (Haug and Smidsrød 1967; Pereira and Cotas 2020). G-block (guluronate) enriched alginate is more rigid than the M-block (mannuronate) enriched alginate as a result of their monomer distributions (Lee and Mooney 2012). Because of its biocompatibility and ability to form a gel in the presence of sodium or calcium, commercial applications of alginate are expanding. Alginate is widely used in the food, cosmetic, paper, agricultural, pharmaceutical, and biomedical industries (Pereira and Cotas 2020). In the healthcare industry, alginate is used as hydrogels in advanced wound care dressings, as an excipient in pills, and as a bio-ink for the 3D printing of tissues and organs (Lee and Mooney 2012). Also, alginate possesses antioxidant and anti-inflammatory properties (Li et al. 2020).

Currently, commercially available alginate is extracted from brown seaweeds (Peteiro 2018). However, ongoing climatic changes and dynamics in the marine ecosystems have affected the yield of brown seaweed alginate. Moreover, alginate extraction from seaweed is cost-intensive, and the harvest process may also harm adjacent seaweed populations. As the demand out-paces conventional alginate production methods, alternative alginate sources and production processes become vital. Two bacterial genera are reported to produce alginates, like *Pseudomonas* and *Azotobacter* (Gawin et al. 2020).

Pseudomonas aeruginosa is a Gram-negative rod-shaped non-fermentative aerobic organism and motile with a monotrichous polar flagellum (Madaha et al. 2020). It is ubiquitous in distribution and can be found in soil, water, and humans. It is an opportunistic pathogen in immunodeficient patients and causes respiratory infections, urinary tract infections, bacteremia, and cystic fibrosis (Driscoll et al. 2007). Alginate is polymerized, modified, and secreted in *P. aeruginosa* by a multiprotein complex that spans the organism's inner and outer membranes (Hay et al. 2013). The proteins that make up this complex are coded by a 12 gene alginate biosynthesis cluster (*algD*, *alg8*, *alg44*, *algK*, *algE*, *algG*, *algX*, *algL*, *algI*, *algJ*, *algF*, and *algA*), in conjunction with another independently localized gene, *algC* (Chitnis and Ohman 1990). The biofilm-forming ability of *P. aeruginosa* protects the cells from external stresses like dehydration, predation, antibiotics, and immune responses (Kostakioti et al. 2013), thereby increasing the pathogenicity of *P. aeruginosa* (Pedersen et al. 1990).

Previously some other *P. aeruginosa* strains were sequenced and analyzed. Pseudomonas PFL-P1 isolated from a marine environment polluted with polycyclic aromatic hydrocarbons (PAH) was investigated for its catabolic conversion mechanisms of phenanthrene (Mahto and Das 2020). Previously, the genome annotations of *P. aeruginosa* UY1PSABAL and UY1PSABAL2 were reported in the context of antibiotic resistance, virulence factors, and other secondary metabolites (Madaha et al. 2020). Srivastava et al. (2020) analyzed the draft genome of two *P. aeruginosa* isolates (VIT PC 7 and VIT PC 9), in the context of the biofilm phenotype, antibiotic resistance, and rhamnolipid production. Similarly, the genome of *P. aeruginosa* KT115 was characterized to understand the conversion of rapeseed oil into di-rhamnolipids. (Liu et al. 2020). However, so far, no genome characterization has been reported focusing on alginate metabolism in *P. aeruginosa*. In this study, we have performed a de novo assembly of the genome of the *P. aeruginosa* strain SW1 isolated from seaweed waste and annotated the genes with a special emphasis on the functional genes involved in alginate metabolism and antibiotic resistance.

Materials and methods

Sample collection and selection of the strain

Degraded seaweeds were collected from the watershed of Kovalam, Chennai, Tamil Nadu, India. The seaweed samples were disintegrated, serially diluted, and cultured on Luria Bertani broth at 37 °C for 48 h. Morphologically distinct mucoid colony growth was selected in the spread plates, and the single colonies were further quadrantally streaked to obtain the pure culture. The pure culture was subjected to a microtiter dish biofilm formation assay (O'Toole 2010). The strain was cultured in trypticase soy broth (TSB) medium 37 °C for 16 h. The diluted culture (1:100) was transferred to a round-bottom 96-well microtiter polystyrene plate (TPP, Trasadingen, Switzerland) and incubated at 37 °C at 48 h and 72 h. After incubation, the cells were removed by washing the plates using deionized distilled water. Subsequently, 100 µL of 0.01% crystal violet (CV) solution was added to all wells containing dry biofilms. After 15 min of incubation, the excess CV was removed by washing with sterile water. Finally, the fixed CV was released by 95% ethanol washing, and 125 µL of the solution was transferred to a new microtiter plate for absorbance detection at 570 nm using a UV spectrophotometer (GENESYS 150 UV–Visible Spectrophotometer, ThermoFisher, India). The sterile medium was used as a blank control.

Phenotypic and biochemical characteristics of the bacterial strain

Morphological and other biochemical characteristics were carried out for the selected strain (Varghese and Joy 2016). Also, the physiological characterization, including the growth at different pH (4–12), temperature (20–50 °C), and salt tolerance (1–5% NaCl), was carried out (Gupta et al. 2013).

MALDI-TOF MS-based bacterial identification

The biochemical characterization of the strain was performed using Matrix-Assisted Laser Desorption–Ionization Time-of-Flight Mass Spectrometry (MALDI-TOF) analysis on a Microflex MALDI-TOF mass spectrometer (Bruker ultrafleXtreme, Bremen, Germany) (Seng et al. 2009). Briefly, purified colonies were spotted onto the MALDI polished plates, and each sample was layered with 2 μL of saturated α-cyano-hydroxycinnamic acid (HCCA matrix made solution) in 50% acetonitrile and 2.5% tri-fluoro acetic-acid and allowed to air dry at room temperature for 5 min. The spectral analysis and matching of sample spectra against the reference database were evaluated using the Bruker Biotyper 3.0 software. The mass spectrometer was calibrated using *E. coli* DH5α as the positive control and a non-inoculated matrix solution as a negative control. MALDI-TOF experiments were performed in triplicates for the strain. All match scores of ≥ 1.9 indicated high-confidence identification, while a score of < 1.7 is unreliable (Angelakis et al. 2011). Normalized MALDI TOF–MS patterns of the strains were compiled using agglomerative hierarchical clustering (AHC).

Molecular identification and phylogenetic analysis

The 16S rRNA gene of the strain was amplified by polymerase chain reaction (PCR) with the universal primers 27F (5′-AGAGTTTGATCCTGGCTCAG-3′) and 1492R (5′-TACGGTTACCTTGTTACGACTT-3′) (Chen et al. 2015). The amplified PCR product was purified (Qiagen PCR product purification kit) and Sanger sequenced (Applied Biosystems SeqStudio Genetic Analyzer, Saint Aubin, France). The sequencing data was analyzed using the Basic Local Alignment Search Tool (BLAST 2.3.0) (https://blast.ncbi.nlm.nih.gov/Blast.cgi). The bacterial strain was identified based on sequence similarity (Altschul et al. 1990). A phylogenetic tree was constructed using MEGA X by the neighbor-joining method (Kumar et al. 2018; Saitou and Nei 1987). *Klebsiella aerogenes* KCTC 2190 was used as the outgroup for phylogenetic analysis.

Alginate assay

Alginate concentration was determined using the carbazole assay (Knutson and Jeanes 1968). The strain was plated on *Pseudomonas* isolation agar (PIA) and incubated at 37°C for 24 h. The bacterial cells were dispersed in phosphate-buffered saline (PBS), and the OD_{600nm} was recorded. The uronic acid content in the buffer was measured and compared with an alginate standard curve made with D-mannuronic acid lactone (Sigma- Aldrich, St. Louis, MO) in the range of 0–100 μg/ mL. The assay colorimetric OD_{600nm} was calibrated to μg uronic acid/mL per $OD_{600nm} = 1$ culture resuspension. The experiment was carried out in triplicates, and the results represented a mean of triplicates with standard deviation.

Genome sequencing of *P. aeruginosa* SW1 strain

Genomic DNA of the strain SW1 was extracted using the QIAamp® DNA Microbiome Kit (Qiagen, Germany) as instructed by the manufacturer. The quantity and quality of isolated DNA were determined using 1% agarose gel and Nano-drop (TECAN-Infinite 200 PRO, Switzerland) by determining the A260/280 ratio. DNA concentration was checked by Qubit® 3.0 Fluorometer (Thermo Scientific Inc, USA) for library preparation. The genomic libraries were created from the DNA aliquots using the Nextera XT Library preparation kit ("XT"; Illumina). Each library pool was loaded and sequenced on a NextSeq 500 device (Illumina Inc, San Diego, CA, USA), which yielded 2 × 150 bp paired-end reads. The final library was analyzed in BioAnalyser 2100 (Agilent Technologies, USA) using the High Sensitivity DNA kit per manufacturer instructions. The paired-end Illumina library with an insert size ranging from 700 to 1500 bp was sequenced using 2 × 150 bp sequencing by synthesis (SBS) chemistry in NextSeq-500. The data were demultiplexed using bcl2fastq2 (version v2.20; Illumina) and used for the genome assembly.

Genome assembly and annotation

The quality of the raw reads was evaluated using FastQC v.0.11.5 (Simon Andrews 2020), and low-quality reads were removed. The adapter regions were trimmed using cutadapt (Saeidipour and Bakhshi 2013). High-quality paired-end short reads of SW1 were assembled with SPAdes (Version: 3.7.1) with default parameters (–careful and –only assembler) (Bankevich et al. 2012). The resulting assembled genome was annotated and analyzed using NCBI Standalone BLAST (Buisine and Chalmers 2004). Assembled genome quality was verified using QUAST (Gurevich et al. 2013) and annotated using Prokka v.2.1.1 (Seemann 2014). Genome annotation was carried out using Rapid Annotation Subsystems Technology (RAST) server v.2.0 (Aziz et al. 2008) and PATRIC v.3.6.2 (Davis et al. 2020).

The gene clusters for secondary metabolites were analyzed with antiSMASH version 5.0 (Blin et al. 2019), and the genomic islands were predicted via IslandViewer 4 (Bertelli et al. 2017). A general overview of the various biological features in the annotated genome was analyzed using RAST. The protein features and antimicrobial resistance genes were annotated using the PATRIC genome analysis server (https://www.patricbrc. org/) (Wattam et al. 2017). The circular genome map and genome-based phylogeny were constructed with the PATRIC. Whole genome similarity was examined using the average nucleotide identity (ANI) calculator and digital DNA hybridization (Yoon et al. 2017).

CAZymes mining and Prediction

Putative genes for carbohydrate active enzymes (CAZymes) encoded in the genome strain of SW1 were classified by dbCAN (using HMMER), CAZy (using DIAMOND), and PPR (using Hotpep) databases (Zhang et al. 2018).

Functional annotation

The functional prediction of protein-coding genes was accomplished by searching against clusters of orthologous groups (COG) via WebMGA server utilizing RSP- BLAST (Wu et al. 2011) and eggNOG (Huerta-Cepas et al. 2017). The whole genome functional annotation was conducted using eggNOG-mapper v2, a tool for functional annotation of large sets of sequences based on fast orthology assignments using precomputed eggNOG v5.0 clusters and phylogenies (Huerta-Cepas et al. 2017). Gene ontology (GO) was analyzed using the Gene Ontology project to provide biologically meaningful gene annotation and their products (http://www.geneontology.org/) (Mi et al. 2021). Metabolic pathways were predicted using KEGG Automatic Annotation Server (KAAS) (Moriya et al. 2007). The graphical map of the circular genome was generated using CGView (http://stothard.afns.ualberta.ca/cgview_server/) (Grant and Stothard 2008).

Data access

The genome sequence of *P. aeruginosa* SW1 was deposited in the SRA database under the accession number PRJNA685877 and in the biosample database with the accession number SAMN17035705 bioproject ID PRJNA685877, respectively. The 16S rDNA gene of the strain SW1 is available at NCBI with accession MW276096.

Results and discussion

Isolation, potential for Biofilm formation and alginate production

A mucoid colony (strain SW1) was isolated from the degraded seaweed, and its potential for biofilm formation was assessed (Table S1). An OD $_{570\ nm} > 0.24$ is consistent with strain SW1 is capable of biofilm formation (Di Domenico et al. 2016). Alginate is one of the extracellular polysaccharides (EPS) present in bacterial biofilms to aid in survival and withstand environmental stress. Strain SW1 achieved 59.3 ± 0.2 µg alginate / mL at the end of 24 h incubation as determined by the carbazole assay, which was substantially higher than levels achieved by a mucoid colony of *P. aeruginosa* PAO579 subjected to the same assay method (Ryan Withers et al. 2013) (Fig. 1). Strain SW1 seems to be a significant alginate producing strain.

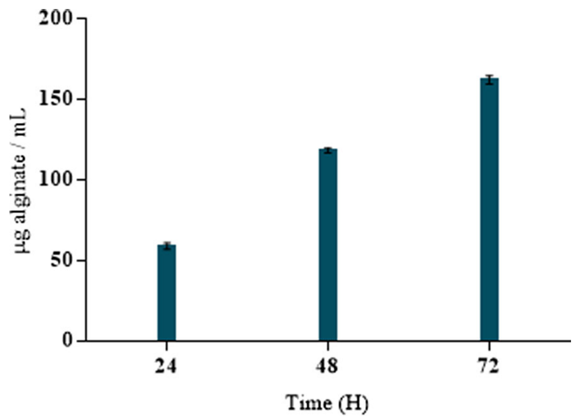

Fig. 1 Graphical representation of alginate production assay at different time intervals

Phenotypic and biochemical characterization

Strain SW1 was identified as a Gram-negative, motile, rod-shaped bacterium (Fig. 2). The strain exhibited oxidase and catalase activity. Assessment for citrate utilization and urea hydrolysis were both positive. The SW1 strain grew in a temperature range between 20°C and 50°C, but the optimal growth after 24 h was observed at 37°C. The isolate displayed growth up to 5% (w/v) NaCl concentration. Although the SW1 strain grew at a wide pH range (4 to 11), maximum growth was observed at pH 7.5. Positive sugar fermentation reactions were observed with D- glucose, fructose, L- arabinose, mannose, D- galactose, and D-xylose, but not D- raffinose, L- rhamnose, and sucrose. Indole production, methyl red, and Voges Proskauer reactions were negative. Positive reactions were observed for the hydrolysis of alginate, casein, starch, cellulose, and gelatin (Table-S2). These identifications match the characteristics of *P. aeruginosa* (Garrity et al. 1991).

MALDI-TOF analysis

Microbial identification based on MALDI-TOF protein fingerprints improves the diagnostic efficiency of infectious diseases by requiring only a few minutes of detection time (Carbonnelle et al. 2012). The raw mass spectrometry profiles were clustered to validate MALDI Biotyper identification accuracy. The representative spectra profile of the bacterial strain SW1 is shown in Fig. S1. MALDI- TOF MS analysis of the strain SW1 matched *P. aeruginosa* in the Bruker Biotyper 3.0 database with a score value of 2.0.

16S rDNA sequencing and phylogenetic analysis

The amplified 16S rRNA gene fragment of 1500 bp was sequenced and subjected to BLAST analysis (Fig. S2). The BLAST analysis revealed that the sequence had the highest similarity to *P. aeruginosa* DSM 50,071 with an identity value of 100%. The phylogenetic analysis of the 16S rRNA gene sequence suggests that the strain belongs to the clade of closely related *P. otitidis* MCC10330 and *P. resinovorans* ATCC 14,235 with 99.93% and 98% sequence similarity, respectively. Other closely related species include *P. stutzeri* ATCC 17,588 (99.9%) and *P. knackmussi* B13 (99%). The strain showed a similar tree topology in the phylogenetic treeing method by

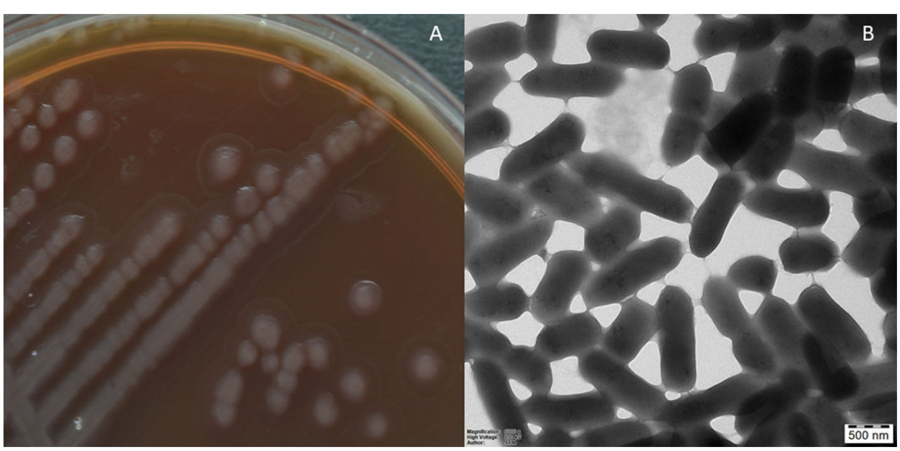

Fig. 2 **a** Colony morphology of mucoid colony; **b** Micrograph of *P. aeruginosa* SW1 using TEM

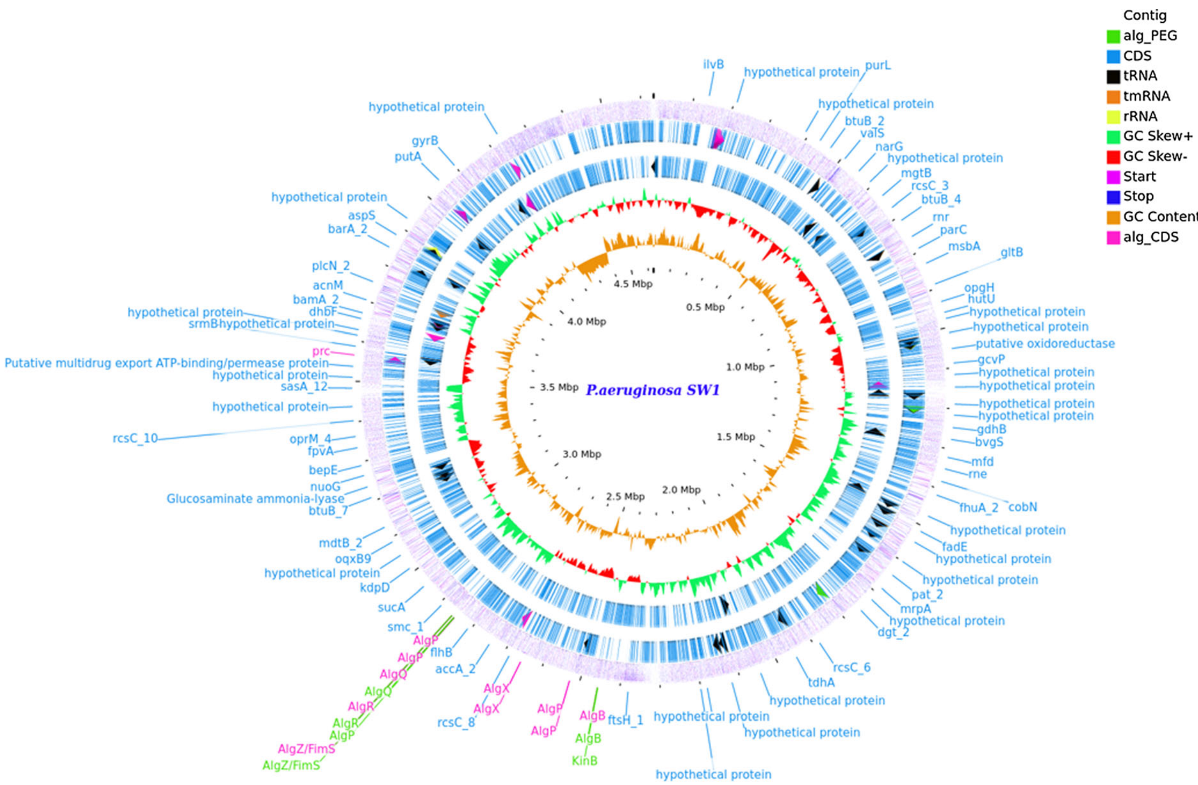

Fig. 3 Circular representation of genome *P. aeruginosa* SW1 using CGview

clustering with *P. aeruginosa* (Fig. S3). Using the neighbor-joining method, the genome and 16S rRNA phylogenetic trees concluded that the strain belongs to *P. aeruginosa*. The partial 16S rRNA gene sequence was submitted to GenBank with the accession number (MW276096).

Genome sequencing, assembly, and annotation

The genome sequencing of *P. aeruginosa* strain SW1 raw reads produced 551,434,596 bases, reduced by quality check and trimming to 461,808,628 bases for 36,87,587 reads. The QUAST assembly resulted in 2,996 contigs. The quality values of N50 and N75 were 21,287 and 7,040 with L50 (169) and L75 (669), respectively. The 2996 contigs cover a total length of 23,258,857 bp, with the largest contig of 600,793 bp with 65.58% G + C content (Table S3). As it was similar to the previously reported genomic studies (Klockgether et al. 2011; Subedi et al. 2018). Analysis of RAST annotated genome using CG view produced a circular genome map illustrating the GC content, tRNA, rRNA, CDS, and contigs (Fig. S4).

PROKKA annotated genome using PATRIC produced a circular genome map (Fig. 3), which identified various gene features, including 6,734 protein-coding sequences (CDS), 8 ribosomal RNA (rRNA), and 63 transfer RNA (tRNA). The protein features consist of 1,636 hypothetical proteins, 5,098 functional assignments, 1,313 EC number assignments, 1,115 GO assignments, and 986 proteins with KEGG pathway assignments (Fig. S5). The genome-based phylogenetic analysis is presented in Fig. S6. The genomic similarity of *P. aeruginosa* SW1 and *P. aeruginosa* PAO1 was found to have an ANI value of 99.37% and dDDH value of 94.90%. The resulting assembled genome was annotated and analyzed using NCBI Standalone BLAST (Table S4). The genome of *P. aeruginosa* strain SW1 consists of 6483 proteins with genus-specific family (Pfam) assignments and 6590 proteins with cross-genus family (PGfam) assignments.

In the whole-genome analysis, "amino acids and derivatives" and "carbohydrates" were the dominant subsystems in *P. aeruginosa* SW1, followed by cofactors, vitamins, prosthetic groups, pigments, cell wall, and capsule, and protein metabolism covering the basic cellular processes. Moreover, subsystems associated with "virulence, disease, and defense" were also present in the SW1 genome; therefore, this strain might be resistant to multiple antibiotics.

CAZymes analysis

The assembled genome of strain SW1 was annotated in the dbCAN meta server for carbohydrate active enzyme domains (CAZymes), as shown in Fig. S7. Considering one annotated protein may contain more than one CAZyme domain, the genome of strain SW1 encompasses a total of 111 annotated proteins with 6 CAZyme domains. An in-depth analysis of the CAZyme profile revealed that several CAZyme domains were related to alginate metabolism. GlycosylTransferase GT1 was distantly related to family GT28, including subunits related to alginate biosynthesis enzymes Alg13 and Alg14 in *Saccharomyces*. Polysaccharide Lyase family PL5_1 includes alginate lyase (EC 4.2.2.3) and endo-β-1,4-glucuronan lyase (EC 4.2.2.14), which play a major role in β elimination of alginate. PL7_1 family includes poly(β-mannuronate) lyase/M-specific alginate lyase (EC 4.2.2.3); α-L-guluronate lyase/G-specific alginate lyase (EC 4.2.2.11); poly-(MG)-lyase / MG-specific alginate lyase (EC 4.2.2.-); endo-β-1,4-glucuronan lyase (EC 4.2.2.14); oligoalginate lyase/exo-alginate lyase (EC 4.2.2.26) which is involved in β elimination of alginate. The functional identification of this breadth of alginate metabolism genes is consistent with the prolific alginate biofilm observed for SW1.

Orthologous genes, GO, and KEGG of strain SW1

Orthologous genes are clusters of homologous genes that are descended from a single ancestral gene. Even after evolution, these genes usually retain functions similar to those of their ancestral genes. We have analyzed orthologous genes through the eggNOG and webMGA (Table S5) for additional functional annotation (Fig. 4). GO annotation classification and KEGG (Kyoto Encyclopedia of Genes and Genomes) metabolic pathway analysis of *P. aeruginosa* SW1 reveals an abundance of subcategories in biological processes, cellular components, molecular function, protein class, and pathway are presented in Fig. S8. The KEGG pathway mapping of target gene candidates was similar to RAST genome annotation with the functional hierarchy of KEGG identifiers and annotation shown in Fig. S9. Notably, 61 genes were associated with "drug resistance", including 11β-lactam resistance genes, three antifolate resistance genes, six vancomycin resistance genes, and 19 cationic antimicrobial peptide resistance genes. These

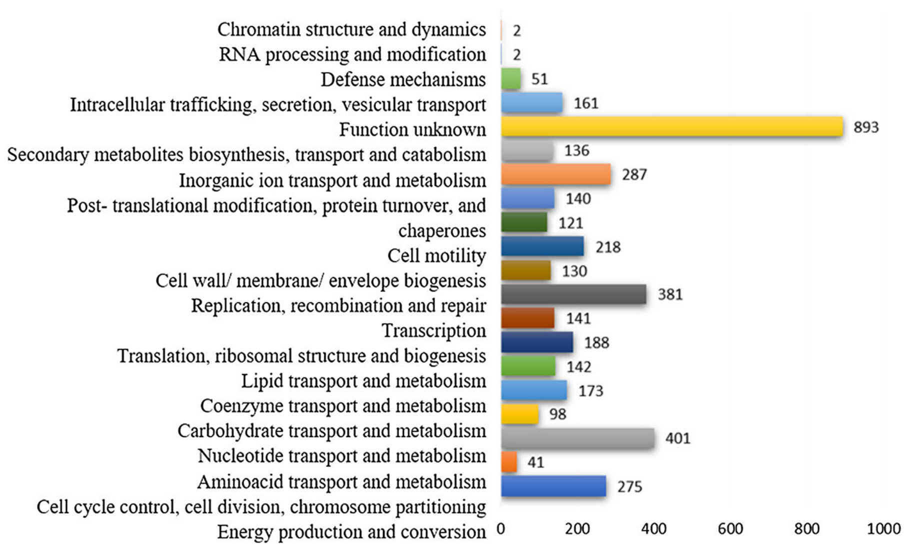

Fig. 4 Cluster of orthologous gene (COG) classification using eggNOG

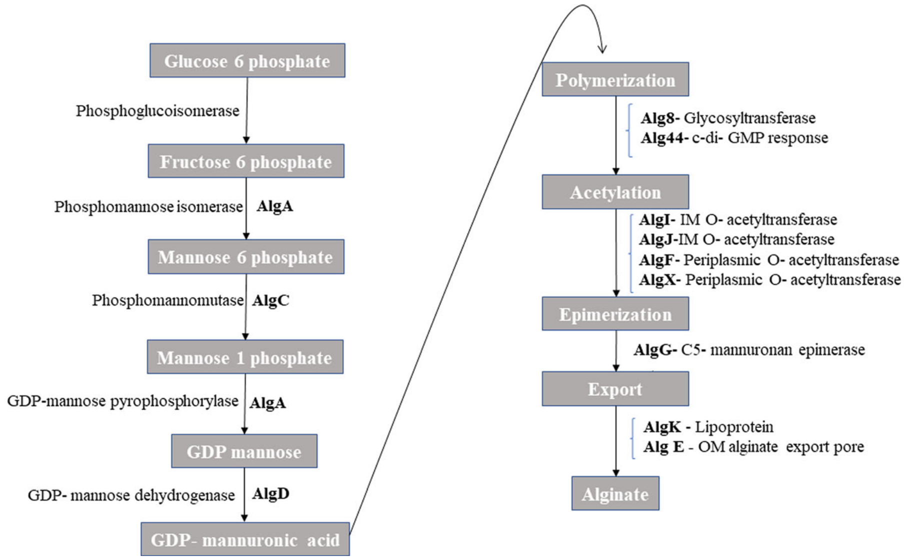

Fig. 5 Bacterial Alginate biosynthesis pathway of *P. aeruginosa*

findings implicate antibiotic resistance in defining the degradative environmental niche of *P. aeruginosa* SW1.

Comparative genomic analysis of COGs and KEGG in five *Pseudomonadales* strains

Pseudomonas aeruginosa SW1 exhibits 3971 clusters of orthologous groups (COGs) and 314 singletons. When compared with COGs of the other five *Pseudomonadales* strains available in the GenBank, a total of 742 COGs were shared by all five strains of *Pseudomonadales* (Fig. S10).

Alginate biosynthesis pathway analysis

Alginate is produced by the synergistic action of a broad spectrum of 'alginate associated enzymes' such as GHs, GTs, CEs, PLs, and related domains of CBMs. The functional annotation of *P. aeruginosa* SW1 was assessed by webMGA and KO (Kegg Orthology) and listed in Tables S6 and S7. Based on Table S6, a total of 19 genes encode proteins related to alginate biosynthesis as an integral component of biofilm formation. Alginate synthesized by *P. aeruginosa* varies depending on environmental conditions and is prominent in biofilm formation's early and late stages. Given the potential biotechnological application of *Pseudomonas* for alginate production, we analyzed the involved genes in more detail below.

Alginate biosynthesis is divided into four different stages: (i) synthesis of precursor substrate in the cytoplasm, (ii) polymerization and cytoplasmic membrane transfer, (iii) periplasmic modification, and (iv) export through the outer membrane (Muhammadi 2007; Hay et al. 2013). The precursor for alginate synthesis fructose-6-phosphate is converted to active precursor GDP-mannuronic acid through a series of enzymatic steps involving the proteins encoded by *algA*, *algC,* and *algD* (Hay et al. 2013). *AlgA* encodes the bifunctional protein phosphate mannose isomerase, which isomerizes fructose-6-phosphate to mannose-6-phosphate. Isomerized mannose-6-phosphate is further converted to mannose-1-phosphate by phosphomannomutase (*algC*). In addition to other exopolysaccharide biosynthetic roles (Franklin et al. 2011), AlgC converts the mannose 6 phosphate into mannose 1 phosphate by the catalysis of GDP mannose phosphorylase activity of AlgA. AlgD encodes GDP-mannose dehydrogenase, which oxidizes GDP-mannose into GDP- mannuronate. GDP mannuronate is the primary substrate for the polymerization of alginate (Hay et al. 2013). The membrane-bound glycosyltransferase Alg8 facilitates polymerization to polymannuronan (Oglesby et al. 2008), with indirect involvement of Alg44, which is involved in secretion by bridging to the periplasmic and outer membrane (Whitney and Howell 2013). The translocation of the alginate polymer also involves a multiprotein scaffold of AlgX, AlgG, and AlgK (Robles-Price et al. 2004). AlgE encodes an anionic selective porin-like outer membrane protein for secretion of mature alginate (Hay et al. 2010).

Numerous other proteins of the alginate COG play various roles in localization, modification, and regulation. AlgL acts as the maintenance protein by degrading misguided alginate trapped in the periplasm and also controls the polymer length (Jain and Ohman 2005). AlgI, AlgJ, and AlgF are involved in the O-acetylation of alginate, where AlgI is a cytoplasmic-membrane protein that transports the acetyl group substrate into the periplasm. AlgK is the lipoprotein that also plays a role in the localization of porin AlgE to the outer membrane (Keiski et al. 2010). AlgK and AlgE interact and share homology to enzymes responsible for the biosynthesis of cellulose, PEL, and poly β-1,6-N-acetyl-D-glucosamine (Rehm 2010). MucD also acts as the essential regulatory protein of the translocation scaffold via AlgX (Hay et al. 2012).

Alginate biosynthesis regulation involves transcriptional and post-transcriptional levels of regulation via single operon genes such as *algU, mucA, mucB, mucC, and mucD*. These operons are homologous to sigma E in *E.coli*. In alginate biosynthesis, AlgU is the analogous sigma 22 factor at the apex of a hierarchy of regulators required for transcription initiation from the AlgD promoter (Ramsey and Wozniak 2005). AlgU is embedded in the inner membrane by the anchored anti-sigma factor MucA (Mathee et al. 1997) and MucB to exhibit the negative regulation of alginate biosynthesis. The degradation of MucA under diverse stresses provides for the protection of *P. aeruginosa*. AlgW encodes a periplasmic protease that cleaves MucA under envelope stress, while MucP encoded intramembrane protease cleaves MucA in its cytosolic side (Damron and Yu 2011). Consistent with its general role in protective biofilm formation, AlgU is also a transcriptional activator for the biosynthesis of other polysaccharides (Ghafoor et al. 2011). Like AlgU, many other genes such as the elements of a 2-component response regulator (AlgR, AlgX, AlgB, and AlgD) are also required for the transcription of the alginate operon, the details of which can be found elsewhere (Baynham et al. 2006) (Merighi et al. 2007). Two positive transcriptional regulators AlgP and AlgQ, are also found in the genome of *P. aeruginosa* SW1. Muhammadi (2007) reported that algP is a histone-like protein that activates the transcription of *algD*, responsible for the synthesis of alginate precursor. AlgQ encoded has been reported to regulate alginate biosynthesis and synthesize virulence factors, neuraminidase, siderophore, rhamnolipid biosurfactant, and extracellular protease (Ledgham et al. 2003). The complexity of the alginate-related proteins and regulatory elements reflects the role of alginate in protective biofilm formation.

Other exopolysaccharides in biofilm formation

Other exopolysaccharides besides alginate are involved in biofilm formation and regulation. The genome of *P. aeruginosa* SW1 includes the PEL exopolysaccharide genes (*PelA through PelG*) for surface adhesion (Vasseur et al. 2005) and PSL exopolysaccharide genes (*PslA, PslB, PslD, PslG,* and *PslL*) which play both a biofilm scaffold role and cellular signaling that enables a non-shared biofilm community resource PSL.

Antimicrobial resistance and efflux gene analysis

Pseudomonas aeruginosa is well known for its antimicrobial resistance (AMR), including antibiotic inactivation and multidrug efflux pumps. Even in the subsystem categorization, 116 gene features were involved in the virulence, disease, and defense subsystem. Strain SW1 carries antibiotic-inactivation enzyme genes, including aminoglycoside phosphotransferases (APH family), chloramphenicol resistance (Cat B family), *Pseudomonas*-derived cephalosporinase family (PDC), and oxacillinase (OXA 50) (Table S8). Aminoglycoside phosphotransferases (APH) lead to the inactivation of antibiotics like kanamycin, neomycin, and streptomycin. The *CatB* gene encodes chloramphenicol acetyltransferase, which confers chloramphenicol resistance (Meng et al. 2020). PDC includes beta-lactamase, which provides resistance against ceftazidime, cefepime, cephalosporin, and carbapenem (Cho et al. 2015). OXA 50 is also a beta-lactamase, which reduces susceptibility to ampicillin, ticarcillin, moxalactam, and meropenem in *P. aeruginosa* (Girlich et al. 2004). These antibiotic resistance genes indicated that Strain SW1 might be a multidrug-resistant bacterium.

Multi efflux pump genes in the strain SW1 annotated genome include those typically associated with substantial antibiotic resistance: MexAB-OprM, MexCD-OprJ, MexEF-OprN, and numerous others (MexHI-OpmD MexJK-OprM/OpmH, MexPQ-OpmE, MexVW-OprM, and MexXY-OMP). Efflux pumps have different substrate specificities, and many factors can regulate their production and activity (e.g., high inocula of bacteria, low pH, and stationary-phase growth) (Scheife 2002). The MexAB-OprM efflux pump overcomes susceptibility to β-lactams, β-lactam antagonists, fluoroquinolones, tetracyclines, tigecycline, novobiocin, and other groups of antibiotics (Li et al. 2015). MexA protein is a lipoprotein that binds with MexB to the porin-like OprM of the outer membrane, promoting one-step efflux of drugs from the cell (Lister et al. 2009). MexCD-OprJ upregulation provides improved tolerance to ciprofloxacin, cefepime, and chloramphenicol. The rates of efflux of these multiple pumps, combined with the outer membrane permeation rates, combine to determine the effectiveness of this exclusion approach to antibiotic resistance.

Conclusion

A mucoid bacterium isolated from decaying seaweed was identified as *Pseudomonas aeruginosa* strain SW1. The identification was confirmed by whole-genome sequencing, assembly, and annotation, MALDI-TOF protein fingerprinting, biochemical/metabolic assays, and 16S rRNA gene sequencing. At least 27 genes involved in alginate and exopolysaccharide biosynthesis were present in the genome. *P. aeruginosa* SW1 displayed high levels of alginate production and biofilm formation. The annotation of this genome, focusing on alginate biosynthesis and antimicrobial resistance genes, could provide a starting point for developing a commercial production strain.

Acknowledgements We would like to acknowledge SRM-DBT Partnership Platform for Contemporary Research Services and Skill Development in Advanced Life Sciences Technologies (Order No. BT/PR12987/INF/22/205/2015), SRM Institute of Science and Technology, Chennai, Tamilnadu, India.

Author contributions RSD performed the experiments and drafted the manuscript. RSD performed isolation, deposition, and identification. WRC and RM performed genome analysis and genome deposition. MR designed all the experiments and supervised the manuscript. MR and PR critically reviewed and revised the manuscript.

Funding Not applicable.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

References

Altschul SF, Gish W, Miller W et al (1990) Basic local alignment search tool. J Mol Biol 215:403–410. https://doi.org/10.1016/S0022-2836(05)80360-2

Andrews S (2020) Babraham bioinformatics - FastQC a quality control tool for high throughput sequence data. Soil 5:47–81

Aziz RK, Bartels D, Best A et al (2008) The RAST server: rapid annotations using subsystems technology. BMC Genomics 9:1–15. https://doi.org/10.1186/1471-2164-9-75

Bankevich A, Nurk S, Antipov D et al (2012) SPAdes: a new genome assembly algorithm and its applications to single-cell sequencing. J Comput Biol 19:455–477. https://doi.org/10.1089/cmb.2012.0021

Baynham PJ, Ramsey DM, Gvozdyev BV et al (2006) The Pseudomonas aeruginosa ribbon-helix-helix DNA-binding protein AlgZ (AmrZ) controls twitching motility and biogenesis of type IV pili. J Bacteriol 188:132–140. https://doi.org/10.1128/JB.188.1.132-140.2006

Bertelli C, Laird MR, Williams KP et al (2017) IslandViewer 4: expanded prediction of genomic islands for larger-scale datasets. Nucleic Acids Res 45:W30–W35. https://doi.org/10.1093/nar/gkx343

Blin K, Shaw S, Steinke K et al (2019) AntiSMASH 5.0: Updates to the secondary metabolite genome mining pipeline. Nucleic Acids Res 47:W81–W87. https://doi.org/10.1093/nar/gkz310

Buisine N, Chalmers R (2004) YBlast, a graphical front end for the standalone BLAST suite. Biotechniques 37:987–989. https://doi.org/10.2144/04376bin01

Carbonnelle E, Grohs P, Jacquier H et al (2012) Robustness of two MALDI-TOF mass spectrometry systems for bacterial identification. J Microbiol Methods 89:133–136. https://doi.org/10.1016/j.mimet.2012.03.003

Chen YL, Lee CC, Lin YL, et al (2015) Obtaining long 16S rDNA sequences using multiple primers and its application on dioxincontaining samples. BMC Bioinform 16:S13. https://doi.org/10.1186/1471-2105-16-S18-S13

Chitnis CE, Ohman DE (1990) Cloning of Pseudomonas aeruginosa algG, which controls alginate structure. J Bacteriol 172:2894–2900. https://doi.org/10.1128/jb.172.6.2894-2900.1990

Cho HH, Kwon GC, Kim S, Koo SH (2015) Distribution of Pseudomonas-derived cephalosporinase and metallo-β-lactamases in carbapenem-resistant Pseudomonas aeruginosa isolates from Korea. J Microbiol Biotechnol 25:1154–1162. https://doi.org/10.4014/jmb.1503.03065

Damron FH, Yu HD (2011) Pseudomonas aeruginosa MucD regulates the alginate pathway through activation of MucA degradation via MucP proteolytic activity. J Bacteriol 193:286–291. https://doi.org/10.1128/JB.01132-10

Davis JJ, Wattam AR, Aziz RK et al (2020) The PATRIC bioinformatics resource center: expanding data and analysis capabilities. Nucleic Acids Res 48:D606–D612. https://doi.org/10.1093/nar/gkz943

Driscoll JA, Brody SL, Kollef MH (2007) The epidemiology, pathogenesis and treatment of Pseudomonas aeruginosa infections. Drugs 67:351–368. https://doi.org/10.2165/00003495-200767030-00003

Franklin MJ, Nivens DE, Weadge JT, Lynne Howell P (2011) Biosynthesis of the pseudomonas aeruginosa extracellular polysaccharides, alginate, Pel, and Psl. Front Microbiol 2:1–16. https://doi.org/10.3389/fmicb.2011.00167

Garrity GM, Bell JA, Lilburn T (2005) "Pasteurellalesord. nov." Bergey's Manual® of Systematic Bacteriology. Springer, Boston, pp 850–912

Gawin A, Tietze L, Aarstad OA et al (2020) Functional characterization of three Azotobacter chroococcum alginate-modifying enzymes related to the Azotobacter vinelandii AlgE mannuronan C-5-epimerase family. Sci Rep 10:1–14. https://doi.org/10.1038/s41598-020-68789-3

Ghafoor A, Hay ID, Rehm BHA (2011) Role of exopolysaccharides in Pseudomonas aeruginosa biofilm formation and architecture. Appl Environ Microbiol 77:5238–5246. https://doi.org/10.1128/AEM.00637-11

Girlich D, Naas T, Nordmann P (2004) Biochemical characterization of the naturally occurring oxacillinase OXA-50 of Pseudomonas aeruginosa. Antimicrob Agents Chemother 48:2043–2048. https://doi.org/10.1128/AAC.48.6.2043-2048.2004

Grant JR, Stothard P (2008) The CGView Server: a comparative genomics tool for circular genomes. Nucleic Acids Res 36:181–184. https://doi.org/10.1093/nar/gkn179

Gupta MK, Shrivastava A, Patel M et al (2013) Optimization of growth parameters for enhanced production of antibacterial compounds by environmentally isolated bacillus optimization of growth parameters for enhanced production of antibacterial compounds by environmentally isolated. Int J Environ Sci 2:11–18

Gurevich A, Saveliev V, Vyahhi N, Tesler G (2013) QUAST: quality assessment tool for genome assemblies. Bioinformatics 29:1072–1075. https://doi.org/10.1093/bioinformatics/btt086

Haug A, Smidsrød O (1967) Strontium-calcium selectivity of alginates [34]. Nature 215:757

Hay ID, Rehman ZU, Rehm BHA (2010) Membrane topology of outer membrane protein AlgE, which is required for alginate production in Pseudomonas aeruginosa. Appl Environ Microbiol 76:1806–1812. https://doi.org/10.1128/AEM.02945-09

Hay ID, Schmidt O, Filitcheva J, Rehm BHA (2012) Identification of a periplasmic AlgK-AlgX-MucD multiprotein complex in Pseudomonas aeruginosa involved in biosynthesis and regulation of alginate. Appl Microbiol Biotechnol 93:215–227. https://doi.org/10.1007/s00253-011-3430-0

Hay ID, Rehman ZU, Moradali MF et al (2013) Microbial alginate production, modification and its applications. Microb Biotechnol 6:637–650. https://doi.org/10.1111/1751-7915.12076

Huerta-Cepas J, Forslund K, Coelho LP et al (2017) Fast genome-wide functional annotation through orthology assignment by eggNOG-mapper. Mol Biol Evol 34:2115–2122. https://doi.org/10.1093/molbev/msx148

Jain S, Ohman DE (2005) Role of an alginate lyase for alginate transport in mucoid Pseudomonas aeruginosa. Infect Immun 73:6429–6436. https://doi.org/10.1128/IAI.73.10.6429-6436.2005

Keiski CL, Harwich M, Jain S et al (2010) AlgK is a TPR-containing protein and the periplasmic component of a novel exopolysaccharide secretin. Structure 18:265–273. https://doi.org/10.1016/j.str.2009.11.015

Klockgether J, Cramer N, Wiehlmann L et al (2011) Pseudomonas aeruginosa genomic structure and diversity. Front Microbiol 2:1–18. https://doi.org/10.3389/fmicb.2011.00150

Knutson CA, Jeanes A (1968) A new modification of the carbazole analysis: application to heteropolysaccharides. Anal Biochem 24:470–481. https://doi.org/10.1016/0003-2697(68)90154-1

Kostakioti M, Hadjifrangiskou M, Hultgren SJ (2013) Bacterial biofilms: development, dispersal, and therapeutic strategies in the dawn of the postantibiotic era. Cold Spring Harb Perspect Med 3:1–23. https://doi.org/10.1101/cshperspect.a010306

Kumar S, Stecher G, Li M et al (2018) MEGA X: molecular evolutionary genetics analysis across computing platforms. Mol Biol Evol 35:1547–1549. https://doi.org/10.1093/molbev/msy096

Ledgham F, Soscia C, Chakrabarty A et al (2003) Global regulation in Pseudomonas aeruginosa: the regulatory protein AlgR2 (AlgQ) acts as a modulator of quorum sensing. Res Microbiol 154:207–213. https://doi.org/10.1016/S0923-2508(03)00024-X

Lee KY, Mooney DJ (2012) Alginate: properties and biomedical applications. Prog Polym Sci 37:106–126. https://doi.org/10.1016/j.progpolymsci.2011.06.003

Li XZ, Plésiat P, Nikaido H (2015) The challenge of efflux-mediated antibiotic resistance in Gram-negative bacteria. Clin Microbiol Rev 28:337–418. https://doi.org/10.1128/CMR.00117-14

Li CW, Li LL, Chen S et al (2020) Antioxidant nanotherapies for the treatment of inflammatory diseases. Front Bioeng Biotechnol 8:1–20. https://doi.org/10.3389/fbioe.2020.00200

Lister PD, Wolter DJ, Hanson ND (2009) Antibacterial-resistant Pseudomonas aeruginosa: clinical impact and complex regulation of chromosomally encoded resistance mechanisms. Clin Microbiol Rev 22:582–610. https://doi.org/10.1128/CMR.00040-09

Liu S, Xu N, Liu H et al (2020) Genome characterization of Pseudomonas aeruginosa KT1115, a high Di-rhamnolipid-producing strain with strong oils metabolizing ability. Curr

Microbiol 77:1890–1895. https://doi.org/10.1007/s00284-020-02009-z
Madaha EL, Mienie C, Gonsu HK et al (2020) Whole-genome sequence of multi-drug resistant Pseudomonas aeruginosa strains UY1PSABAL and UY1PSABAL2 isolated from human broncho-alveolar lavage, Yaoundé, Cameroon. PLoS ONE 15:1–16. https://doi.org/10.1371/journal.pone.0238390
Mahto KU, Das S (2020) Whole genome characterization and phenanthrene catabolic pathway of a biofilm forming marine bacterium Pseudomonas aeruginosa PFL-P1. Ecotoxicol Environ Saf 206:111087. https://doi.org/10.1016/j.ecoenv.2020.111087
Mathee K, McPherson CJ, Ohman DE (1997) Posttranslational control of the algT (algU)-encoded σ22 for expression of the alginate regulon in Pseudomonas aeruginosa and localization of its antagonist proteins MucA and MucB (AlgN). J Bacteriol 179:3711–3720. https://doi.org/10.1128/jb.179.11.3711-3720.1997
Meng J, Liu Y, Xie Z et al (2020) Nucleus distribution of cathepsin B in senescent microglia promotes brain aging through degradation of sirtuins. Neurobiol Aging 96:255–266. https://doi.org/10.1016/j.neurobiolaging.2020.09.001
Merighi M, Lee VT, Hyodo M et al (2007) The second messenger bis-(3′-5′)-cyclic-GMP and its PilZ domain-containing receptor Alg44 are required for alginate biosynthesis in Pseudomonas aeruginosa. Mol Microbiol 65:876–895. https://doi.org/10.1111/j.1365-2958.2007.05817.x
Mi H, Ebert D, Muruganujan A et al (2021) PANTHER version 16: a revised family classification, tree-based classification tool, enhancer regions and extensive API. Nucleic Acids Res 49:D394–D403. https://doi.org/10.1093/nar/gkaa1106
Moriya Y, Itoh M, Okuda S et al (2007) KAAS: an automatic genome annotation and pathway reconstruction server. Nucleic Acids Res 35:182–185. https://doi.org/10.1093/nar/gkm321
Muhammadi AN (2007) Genetics of bacterial alginate: alginate genes distribution, organization and biosynthesis in bacteria. Curr Genomics 8:191–202. https://doi.org/10.2174/138920207780833810
O'Toole GA (2010) Microtiter dish biofilm formation assay. J vis Exp. https://doi.org/10.3791/2437
Oglesby LL, Jain S, Ohman DE (2008) Membrane topology and roles of Pseudomonas aeruginosa Alg8 and Alg44 in alginate polymerization. Microbiology 154:1605–1615. https://doi.org/10.1099/mic.0.2007/015305-0
Pedersen SS, Kharazmi A, Espersen F, Hoiby N (1990) Pseudomonas aeruginosa alginate in cystic fibrosis sputum and the inflammatory response. Infect Immun 58:3363–3368. https://doi.org/10.1128/iai.58.10.3363-3368.1990
Pereira L, Cotas J (2020) Introductory chapter: alginates-a general overview. Alginates Recent Uses Nat Polym. https://doi.org/10.5772/intechopen.88381
Peteiro C (2018) Alginate production from marine macroalgae, with emphasis on kelp farming. In: Alginates and their-biomedical applications. Springer, Singapore, pp 27–66
Ramsey DM, Wozniak DJ (2005) Understanding the control of Pseudomonas aeruginosa alginate synthesis and the prospects for management of chronic infections in cystic fibrosis. Mol Microbiol 56:309–322. https://doi.org/10.1111/j.1365-2958.2005.04552.x
Rehm BHA (2010) Bacterial polymers: biosynthesis, modifications and applications. Nat Rev Microbiol 8:578–592. https://doi.org/10.1038/nrmicro2354
Robles-Price A, Wong TY, Sletta H et al (2004) AlgX is a periplasmic protein required for alginate biosynthesis in Pseudomonas aeruginosa. J Bacteriol 186:7369–7377. https://doi.org/10.1128/JB.186.21.7369-7377.2004
Ryan Withers T, Heath Damron F, Yin Y, Yu HD (2013) Truncation of type IV pilin induces mucoidy in Pseudomonas aeruginosa strain PAO579. Microbiologyopen 2:459–470. https://doi.org/10.1002/mbo3.86
Saeidipour B, Bakhshi S (2013) The relationship between organizational culture and knowledge management, & their simultaneous effects on customer relation management. Adv Environ Biol 7:2803–2809
Saitou N, Nei M (1987) ESCALA CIWA-AR Escala CIWA-Ar(Clinical institute withdrawal assesment for alcohol) Evaluación del Síndrome de Abstinencia Alcohólica. Mol Biol Evol 4:406–425
Scheife RT (2002) Pharmacotherapy, The Journal of Human Pharmacology and Drug Therapy Encycl Clin Pharm 724–725. https://doi.org/10.3109/9780824706081.125
Seemann T (2014) Prokka: rapid prokaryotic genome annotation. Bioinformatics 30:2068–2069. https://doi.org/10.1093/bioinformatics/btu153
Seng P, Drancourt M, Gouriet F et al (2009) Ongoing revolution in bacteriology: routine identification of bacteria by matrix-assisted laser desorption ionization time-of-flight mass spectrometry. Clin Infect Dis 49:543–551. https://doi.org/10.1086/600885
Srivastava P, Gomathinayagam S, Easwaran N et al (2020) Comparative data analsysis of two multi-drug resistant homoserine lactone and rhamnolipid producing Pseudomonas aeruginosa from diabetic foot infected patient. Data Br. https://doi.org/10.1016/j.dib.2020.106071
Subedi D, Vijay AK, Kohli GS et al (2018) Comparative genomics of clinical strains of Pseudomonas aeruginosa strains isolated from different geographic sites. Sci Rep 8:1–14. https://doi.org/10.1038/s41598-018-34020-7
Varghese N, Joy PP (2016). Microbiological laboratory Manual. Kerala agricultural University, Pineapple research station, Vazhakulam, Muvattapuzha
Vasseur P, Vallet-Gely I, Soscia C et al (2005) The pel genes of the Pseudomonas aeruginosa PAK strain are involved at early and late stages of biofilm formation. Microbiology 151:985–997. https://doi.org/10.1099/mic.0.27410-0
Wattam AR, Davis JJ, Assaf R et al (2017) Improvements to PATRIC, the all-bacterial bioinformatics database and analysis resource center. Nucleic Acids Res 45:D535–D542. https://doi.org/10.1093/nar/gkw1017
Whitney JC, Howell PL (2013) Synthase-dependent exopolysaccharide secretion in gram-negative bacteria. Trends Microbiol 21:63–72. https://doi.org/10.1016/j.tim.2012.10.001
Wu S, Zhu Z, Fu L et al (2011) WebMGA: a customizable web server for fast metagenomic sequence analysis. BMC Genomics. https://doi.org/10.1186/1471-2164-12-444
Yoon SH, S-min Ha, Lim J et al (2017) A large-scale evaluation of algorithms to calculate average nucleotide identity.

Antonie van Leeuwenhoek, Int J Gen Mol Microbiol 110:1281–1286. https://doi.org/10.1007/s10482-017-0844-4

Zhang H, Yohe T, Huang L et al (2018) DbCAN2: a meta server for automated carbohydrate-active enzyme annotation. Nucleic Acids Res 46:W95–W101. https://doi.org/10.1093/nar/gky418